Using Transformers to Reconstruct Dark Matter Events at the DELight Experiment

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Teaching a Neural Network to see Dark Matter

About 85% of the matter in the universe has never been directly detected. Its gravity shapes galaxies, but it has slipped past every particle detector ever built. For decades, the leading candidates were heavy particles that barely interact with ordinary matter, but repeated experiments have failed to detect them. This has turned attention to lighter dark matter.

Particle masses are often quoted as energies in electronvolts, using $E = mc^2$. The DELight experiment is designed to detect dark matter with masses down to 100 MeV/c^2—nearly ten times lighter than a proton. If one of these particles collides with ordinary matter, it only transfers a few tens of electronvolts, roughly ten trillion times less energy than a single raindrop hitting the ground.

Catching a signal that faint requires an extraordinarily sensitive detector. DELight, planned for construction in Heidelberg, Germany, is a cell of superfluid helium (a liquid without friction) about the size of a dinner plate. Held at around 20 mK, it is designed to detect energy transfers down to 20 eV (well below what a 100 MeV/c^2 recoil deposits). As shown in Figure 1, when a dark matter particle strikes a helium nucleus, the recoil produces three signals: a prompt flash of ultraviolet light, an evaporated helium atom, and an excited helium molecule that decays slowly, releasing light long after the recoil. All three deposit energy as a temperature change on an array of 56 quantum sensors, each sensitive to changes of a few millionths of a degree.

The main challenge is turning the 56 faint signals back into a physical event: the energy released in the recoil, where it happened, and whether it came from dark matter or a background electron. The standard method, called optimum filtering, extracts a best-fit energy amplitude from each sensor independently. However, looking at each sensor alone misses the bigger picture, since the strongest sign of a dark matter event comes from how all 56 sensors behave together.

This project, DELiT, takes a different approach. It uses a transformer, the same kind of neural network behind tools like ChatGPT, to learn which sensors matter most for each event using a mechanism called attention. Instead of treating each sensor separately, the network uses the detector geometry to identify patterns across all 56 sensors at once, learning how their timing and signal shape relate to the underlying event. It performs three tasks in parallel: estimating the recoil energy, locating where in the helium the event happened, and deciding whether it was a dark matter recoil or a background electron. Figure 1 shows the full pipeline.

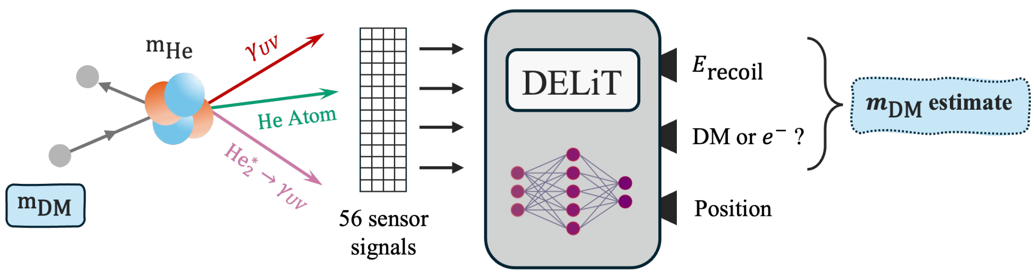

Figure 1: The full DELiT pipeline, from dark matter recoil to inferred dark matter mass.

By combining all 56 sensors, DELiT reliably reconstructs events down to 50 eV with 21% energy resolution and no contamination from electron backgrounds. This is above DELight's 20 eV target but far below previous searches. Since a smaller recoil energy corresponds to a smaller particle mass, each step lower in threshold opens up a new mass range. By making DELight's data usable for a dark matter search, and scaling to the next generation of helium-based detectors with larger arrays and lower thresholds, this approach could bring us closer to answering one of physics' biggest questions: what is most of the universe actually made of?

Abstract

The DELight experiment is a planned direct-detection experiment for light dark matter, using superfluid ^{4}He instrumented with 56 magnetic microcalorimeters in a dual-layer geometry. It is designed to measure energy deposits down to 20 eV, probing dark matter masses below 100 MeV. Reaching this sensitivity requires accurate event reconstruction from the full 56-channel waveform data, beyond the per-channel optimum filter used in single-sensor microcalorimetry. This work presents DELiT, a transformer architecture trained on simulated waveforms for joint reconstruction of recoil energy, nuclear-versus-electronic recoil type, and (x, y) interaction position. Attention between sensor channels is biased by an iteratively updated pairwise representation initialised from the detector geometry, and per-channel optimum-filter amplitudes are injected residually. On 300,000 simulated events DELiT attains, at 50 eV, an energy resolution $\sigma_E/E = 20.9 \pm 0.3\%$, position RMSE of 4.57 ± 0.04 mm—well below the 43.87 mm minimum sensor spacing—and zero electronic-recoil leakage. An ablation isolates the pair update with geometric features as the operative mechanism, jointly reducing σ_E/E by 20% and position RMSE by 16% at 50 eV. Adopting 50 eV as the operating threshold ($E/\sigma_E = 4.78 \pm 0.07$) yields a dark matter mass reach $m_\chi \geq 121$ MeV via elastic-scattering kinematics, the lowest projected reach for any superfluid-helium dark matter detector.

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1 Introduction

1.1 The Search for Light Dark Matter

Dark matter, a fundamental component of the universe, accounts for about five times more mass-energy than baryonic matter [1]. Its existence is inferred from galactic rotation curves [2], cosmic microwave background anisotropies [1], and large-scale structure observations [3]. Despite this empirical evidence, its particle nature remains unknown.

Direct detection experiments have historically targeted weakly interacting massive particles (WIMPs) in the GeV–TeV mass range. However, decades of null results have excluded significant portions of this parameter space. The LUX-ZEPLIN experiment currently provides the most stringent limits above approximately 9 GeV [4]. These constraints and theoretical developments in hidden-sector and dark-photon models have redirected focus toward lighter dark matter candidates below the GeV scale [5, 6].

The detection of light dark matter introduces fundamentally different challenges. When the dark matter mass falls below that of the target nucleus, the maximum recoil energy from elastic scattering decreases rapidly. For conventional heavy targets such as xenon or germanium, the resulting recoil energies often fall below detectable thresholds.

This limitation motivates both the use of lighter target materials and the development of detectors sensitive to energy deposits at the eV scale. Achieving these low thresholds is essential for probing the sub-GeV dark matter parameter space and represents the primary experimental challenge in the search for light dark matter.

1.2 The DELight Experiment

Superfluid ^{4}He represents a promising target material for light dark matter detection. Its low nuclear mass facilitates efficient kinematic energy transfer from sub-GeV dark matter, while the absence of long-lived radioisotopes and the freezing out of other atomic species at superfluid temperatures render it an intrinsically radiopure target.

Energy deposition in superfluid helium generates three signals: quasiparticles (collective excitations of the superfluid), UV scintillation, and long-lived excited helium molecules (excimers) [7]. Scattering of a dark matter particle with a helium nucleus results in a nuclear recoil (NR), whereas electron recoils (ERs) arise from gamma and beta interactions with atomic electrons, the dominant background. The energy distribution among these signals differs between NRs and ERs, with NRs yielding a larger quasiparticle component through enhanced elastic scattering [7]. This distinction enables discrimination between dark matter interactions and background events. The Direct search Experiment for Light dark matter (DELight) is specifically designed to exploit these differences (Figure 1.1).

DELight will employ magnetic microcalorimeters (MMCs), which convert energy depositions into voltage signals by measuring the temperature-dependent magnetisation of a paramagnetic sensor [8]. MMCs have demonstrated a single-sensor energy resolution of 1.25 eV [9], enabling a projected trigger threshold of 20 eV for the full 56-channel array [10].

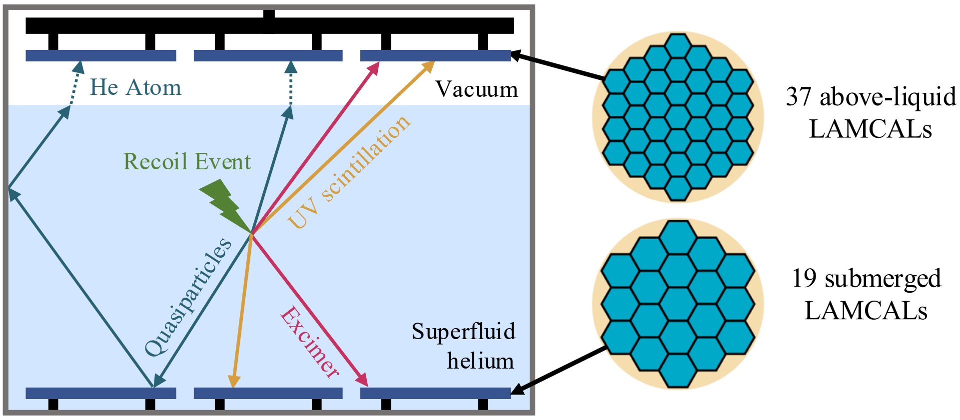

Figure 1.1: DELight's phase-I dual-layer geometry, 30 cm in diameter with a 15 mm vertical separation between sensor layers. The superfluid helium target is instrumented with 56 magnetic microcalorimeter (MMC) based sensors (LAMCALs). The 37 above-liquid sensors detect all three signal channels, including quasiparticle signals via quantum evaporation of helium atoms across the liquid surface [7]. The 19 submerged sensors detect only scintillation and excimer signals. This asymmetry provides enhanced sensitivity to the quasiparticle-to-scintillation ratio that distinguishes nuclear from electron recoils.

As depicted in Figure 1.2, this extends DELight's sensitivity to dark matter masses below 100 MeV, surpassing the reach of existing helium-based detectors. The closest comparable experiment is HeRALD, which also targets superfluid helium signal channels but employs transition edge sensors. HeRALD has demonstrated quasiparticle detection with a single prototype calorimeter, achieving an energy threshold of 145 eV, limiting sensitivity to masses above approximately 220 MeV [11].

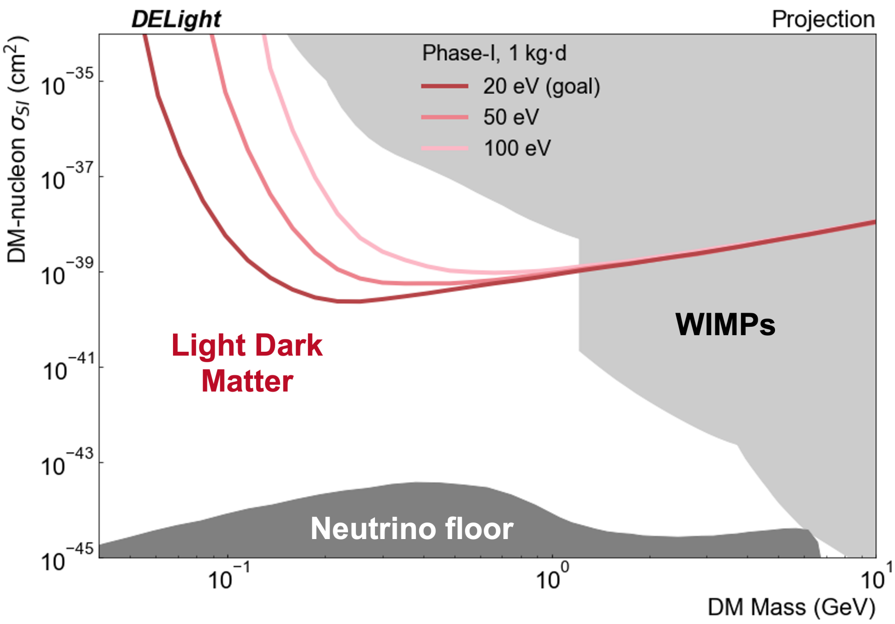

Figure 1.2: DELight's projected sensitivity to spin-independent dark matter–nucleon scattering cross section as a function of dark matter mass. Shown for a 1 kg·day exposure at projected energy thresholds of 20 eV (goal), 50 eV, and 100 eV. The shaded regions indicate masses and cross sections excluded by previous WIMP searches and by the irreducible background from coherent neutrino scattering in helium. Reproduced from [12].

1.3 Event Reconstruction and Project Aims

To achieve its projected sensitivity, DELight must accurately reconstruct three key quantities from an event: the deposited energy, recoil type, and interaction position. Accurate energy reconstruction determines the effective analysis threshold—the lowest energy at which signals can be distinguished from noise with high confidence—and therefore the accessible dark matter mass range (Figure 1.2). Recoil classification suppresses the dominant electromagnetic background, while position reconstruction identifies and rejects surface events, further reducing contamination.

In microcalorimeter experiments, the standard approach to energy reconstruction is the optimum filter (OF). It is a frequency-domain matched filter that estimates pulse amplitudes by minimising a chi-squared statistic with respect to a known signal template [9]. However, the OF operates on individual sensor traces independently, which makes it suboptimal for DELight's 56-channel array. Furthermore, while recoil classification can be performed downstream using OF-derived quantities such as per-channel amplitude ratios, it is not intrinsic to the filter, and position reconstruction lies entirely outside its scope. As a result, the OF addresses only part of the problem and does not leverage the full multi-channel information available.

This motivates the use of machine learning, which can jointly process all sensor channels. By learning temporal, amplitude, and geometric correlations, machine learning may improve energy estimation beyond the per-channel OF. It also naturally frames energy reconstruction, recoil classification, and position reconstruction as a multi-task learning problem, where shared patterns in the data may benefit all three objectives.

Building on this motivation, this project has three aims: first, to develop DELiT, a transformer-based architecture for simultaneous reconstruction of event energy, recoil type, and interaction position from DELight's 56-channel simulated waveform data; second, to evaluate whether incorporating physics-informed architectural priors, encoding detector geometry and signal structure, improves reconstruction performance through a systematic ablation study; and third, to establish the effective analysis threshold and map this to a minimum detectable dark matter mass.

2 Background

2.1 Signal Formation in Superfluid Helium

Energy deposition in superfluid helium depends on recoil type. In a nuclear recoil, an incoming particle, such as a dark matter candidate or neutron, scatters elastically off a helium nucleus, which recoils through the liquid. An electron recoil occurs when gamma or beta interactions transfer energy to an atomic electron. ERs dominate the background due to environmental radioactivity, making NR/ER discrimination essential for signal identification. In both NRs and ERs, energy is dissipated through three channels: elastic scattering, electronic excitation, and ionisation. The complete set of pathways is shown in Figure 2.1.

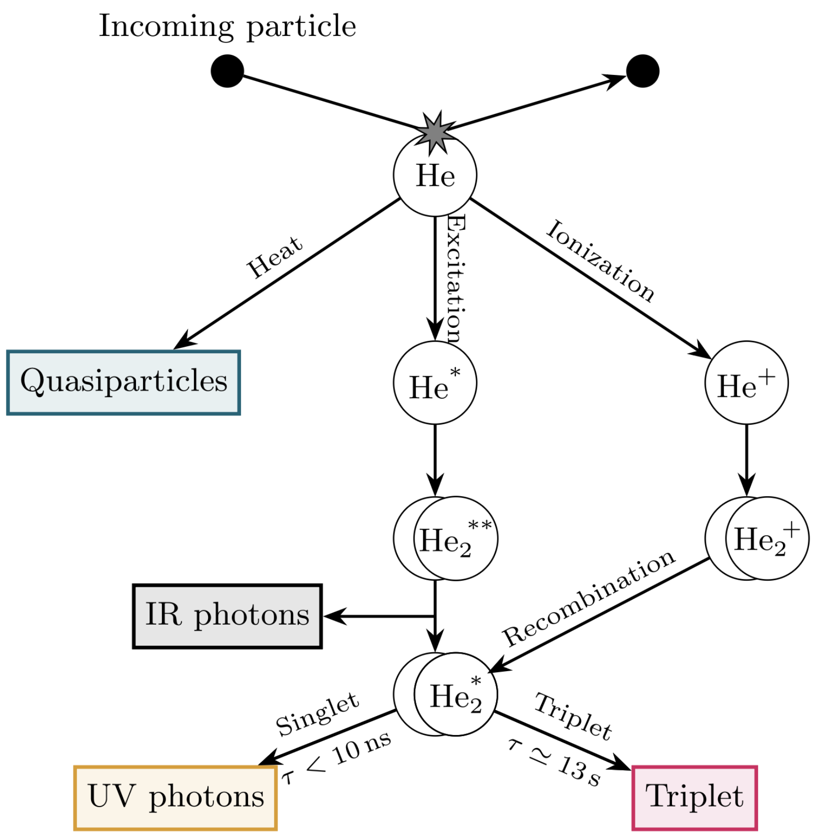

Figure 2.1: Signal formation pathways following energy deposition in superfluid ^{4}He [7].

Elastic scattering produces quasiparticles—phonons and rotons—which propagate ballistically through the superfluid. At millikelvin temperatures, their mean free paths exceed the detector dimensions, allowing them to reach the liquid's boundaries without significant scattering. At the liquid–vacuum interface, quasiparticles with energy exceeding the quantum evaporation threshold of 0.62 meV eject helium atoms into the vacuum [7]. These evaporated atoms are adsorbed by DELight's above-liquid sensors, producing a direct measure of the quasiparticle signal. This signal channel is exclusive to the 37 sensors above the liquid.

Electronic excitation and ionisation generate photonic and molecular signals. Both processes require a minimum energy of 19.82 eV to promote a helium atom from its ground state [13]; below this threshold, only elastic scattering occurs. Excited helium atoms rapidly decay (within 1 μs) to metastable singlet (2^1S) or triplet (2^3S) states, emitting infrared photons [7]. These photons are not detected separately in DELight phase-I because the sapphire LAMCAL substrates are transparent at infrared wavelengths. The metastable atom binds with a neighbouring ground-state atom to form an excimer, a short-lived diatomic molecule bound in the excited state. Ionisation contributes similarly: recombination of thermalised electrons with He_2^+ ions produces both singlet and triplet excimers [7].

Singlet excimers decay within nanoseconds and emit ∼15.4 eV UV photons [7]. The liquid is transparent at this energy, which allows isotropic propagation to all surrounding sensors. In contrast, triplet excimers have a ∼13 s bulk lifetime due to spin-forbidden decay [7]. They travel ballistically (2–4 m/s) and, upon reaching a sensor surface, quench via electron interactions, depositing 17.8 eV on millisecond timescales. This produces a temporally distinct signal.

The energy partition across signal channels depends on both recoil type and energy, as shown in Figure 2.2. For ERs above a few hundred eV, the distribution is roughly constant: ∼32% quasiparticles, ∼33% UV, ∼24% triplets, and ∼11% IR [7]. In contrast, NRs lose a larger fraction of energy through elastic scattering, increasing the relative quasiparticle contribution [7]. The difference in the quasiparticle-to-scintillation ratio provides the primary method for recoil discrimination.

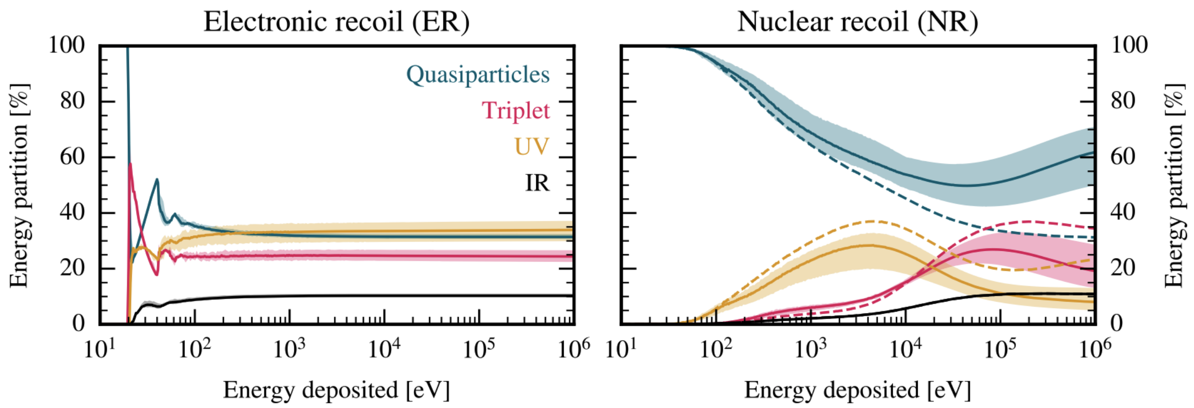

Figure 2.2: Partitioning of electron (left) and nuclear (right) recoil energy into quasiparticles, triplet excimers, UV scintillation, and IR photons in superfluid helium. Coloured bands indicate systematic uncertainties from cross-section measurements, and dashed lines in the right panel show the partitioning prior to Penning quenching [7].

Near the 19.82 eV excitation threshold, the discrimination power collapses. Below this energy, neither NRs nor ERs can excite helium atoms, so both produce only quasiparticles and become indistinguishable by signal composition. Signals in this regime are weak and confined to a small subset of channels, making near-threshold reconstruction the central challenge addressed in this work.

2.2 Simulation and Dataset

The dataset is fully simulated, as DELight remains in the design phase and no experimental event data is available. The simulation proceeds in three stages: (i) energy partitioning, (ii) waveform synthesis, and (iii) noise injection.

For a given recoil type and energy, DELight's Monte Carlo model propagates the recoiling particle through liquid helium [7]. It samples elastic scattering, excitation, and ionisation using measured cross sections. The particle is tracked until its kinetic energy drops below the 19.82 eV excitation threshold. Below this level, only elastic scattering occurs, and all the remaining energy is converted into quasiparticles. The output is a set of per-event energy fractions distributed among the signal channels described in §2.1.

These fractions are mapped to time-domain waveforms for each of the 56 sensors using two templates [9]. A quasiparticle evaporation template captures the above-liquid sensor response, while a scintillation template is used for both prompt UV photons and triplet excimers, whose surface quenching produces a similar calorimetric pulse shape at a later time. Pulse amplitude and arrival time depend on the interaction position relative to each sensor. The spatial pattern of signals encodes the event position across the detector.

Noise is generated independently for each sensor using the power spectral density (PSD) measured from an MMC device during ^{55}Fe calibration [9]. The PSD, $J(\nu)$, characterises additive baseline noise and is independent of signal amplitude. Consequently, a PSD derived from measurements at the 5.9 keV calibration energy is also applicable to the lower-signal energies considered here. It is obtained from baseline data by Fourier transforming noise-only traces recorded prior to signal arrival. For each event and each sensor channel, complex Fourier coefficients are sampled from a standard normal distribution and scaled by $\sqrt{J(\nu)}$. This produces independent noise realisations per sensor with the correct frequency structure. An inverse Fourier transform returns the noise to the time domain, where it is added to the signal trace. The procedure assumes stationary Gaussian noise over the acquisition window, which is well satisfied for individual traces at millisecond timescales.

The dataset contains 300,000 events distributed equally across six recoil energies:
(10, 20, 50, 100, 200, and 500 eV) with a balanced 50:50 NR/ER split at each energy. This balance is a training choice; in reality, ERs dominate by orders of magnitude. Training on balanced classes ensures the model learns to identify NR events rather than defaulting to the majority class. The evaluation metrics used in this work are chosen to reflect performance under realistic class imbalance. The discrete energies span the region of greatest relevance to DELight sensitivity. However, they do not represent a physical recoil spectrum, which would be continuous.

Each event is a $56 \times 65{,}536$ array: one time-domain trace per sensor, sampled at 3.9 MHz over a $\sim$16.8 ms acquisition window. Interaction vertices are sampled uniformly in three dimensions across the helium volume (Figure 1.1), consistent with a constant target number density in the homogeneous liquid. The planar sensor geometry provides no instrumentation on the side walls, so position reconstruction is restricted to (x, y). Example waveforms for NR events at 20 eV and 500 eV, shown in Figure 2.3, illustrate the transition from a clear signal to noise-dominated.

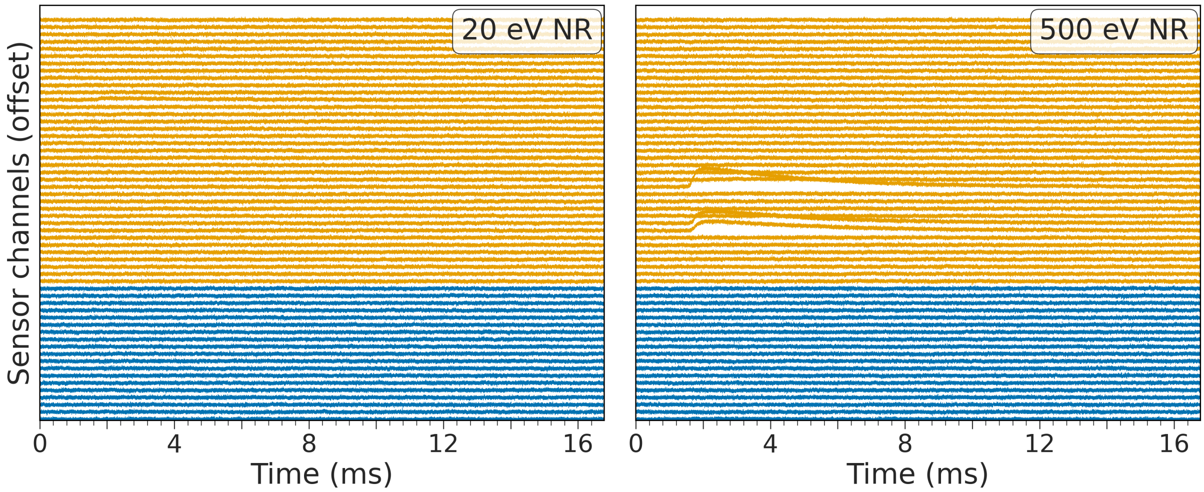

Figure 2.3: Simulated 56-channel waveforms for nuclear recoil events at 20 eV (left) and 500 eV (right). Each trace is vertically offset for visibility. Orange traces correspond to the 37 above-liquid sensors and blue traces to the 19 submerged sensors. At 500 eV, clear pulses are visible in several above-liquid channels, while at 20 eV, no signal is discernible above the noise floor in any channel.

2.3 The Optimum Filter

The optimum filter (OF) is the standard method for energy reconstruction in microcalorimeter experiments [14, 15]. It estimates pulse amplitudes from noisy traces by matched filtering in the frequency domain using a known signal template. In DELight, the OF is applied to each of the 56 sensor traces using the layer-specific templates described in §2.2.

The filter minimises a chi-squared statistic weighted by the noise power spectral density $J(\nu)$, returning an amplitude estimator

$$\hat{a} = \frac{\sum_{\nu} S^{*}(\nu)\, A(\nu)/J(\nu)}{\sum_{\nu} |A(\nu)|^{2}/J(\nu)}, \tag{2.1}$$

where $S(\nu)$ is the measured trace and $A(\nu)$ the signal template. A time shift is introduced via a phase factor in $A(\nu)$, with the optimal delay selected by maximising $\hat{a}$ [9]. The filter returns, for each channel, an amplitude $\hat{a}$ and a goodness-of-fit χ^2.

For stationary Gaussian noise and a known template, the OF is the minimum-variance linear unbiased estimator of the pulse amplitude, with resolution depending only on the template and noise spectrum. Under these conditions, single-sensor measurements using ^{55}Fe calibration data achieve $\Delta E_{\mathrm{FWHM}} = 1.25 \pm 0.17$ eV [9], demonstrating that eV-scale resolution is achievable when the OF assumptions are well satisfied.

In DELight, these conditions are not met. Each above-liquid sensor records a superposition of quasiparticle, scintillation, and triplet contributions, each with distinct timescales. The OF fits a single quasiparticle template to this composite signal, introducing a systematic shape mismatch. In addition, low-energy events distribute their energy across multiple sensors, leaving each channel with only a small fraction of the total deposit and a reduced signal-to-noise ratio. The OF processes each channel independently, discarding correlations in amplitude, timing, and spatial structure across the array. It does not provide position reconstruction or NR/ER classification.

2.4 Machine Learning for Event Reconstruction

Machine learning models can learn structure across time, sensors, and detector geometry that per-channel methods cannot [16, 17]. Existing architectures in high-energy physics address parts of this problem, but none fully match DELight's requirements [5].

Convolutional neural networks (CNNs) perform well on regular grids and have improved energy reconstruction and classification in calorimetry and direct detection experiments [18, 19], but DELight's irregular sensor layout cannot be mapped to a grid without degrading spatial information. Recurrent neural networks (RNNs) model sequential structure and have been applied to time-series classification in particle physics [16], but scale poorly to long, multi-channel waveforms and do not naturally encode spatial relationships between sensors. Graph neural networks (GNNs) address irregular geometries and have achieved strong performance in tracking and calorimetry [20, 21], but standard GNNs do not explicitly model temporal waveform evolution, which is essential for pulse-shape-based discrimination; temporal extensions exist but remain an active area of development [22, 23].

DELight therefore requires an architecture that can model both long-range temporal structure within each waveform and spatial correlations across an irregular sensor array. Transformers achieve this through self-attention [24], a mechanism that allows each element of the input to directly compare itself with every other element and determine which information is most relevant. Self-attention has shown strong performance in jet classification for collider physics [25] and unlike CNNs, which compute local interactions, or RNNs, which process elements sequentially, it computes all pairwise comparisons in a single operation. Each element is projected into three learned vectors: a query Q (what information it seeks), a key K (what information it contains), and a value V (the information it contributes to the output). The attention operation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.2}$$

where $QK^\top$ scores how relevant one element is to another, and the softmax converts these scores into weights. The scaling factor $\sqrt{d_k}$, where d_k is the key vector dimension, prevents large dot products from saturating the softmax. The output is a weighted sum of all values, so each element becomes a context-dependent combination of the entire input.

Applied to DELight's waveform data, this mechanism operates across two axes. Temporally, attention links waveform features such as rise time, pulse width, and triplet quenching delays. Spatially, it relates signals across neighbouring channels to infer interaction position from amplitude and timing patterns. Multi-head attention learns multiple such relationships in parallel, enabling simultaneous sensitivity within each axis. This suits DELight's reconstruction problem, where energy estimation, recoil classification, and position reconstruction all depend on the same underlying signal structure.

3 Methods

This chapter presents DELiT (DELight interaction Transformer), a transformer-based model for joint energy regression, NR/ER classification, and (x, y) position reconstruction from 56-channel simulated waveform data. Figure 3.1 illustrates the complete pipeline. The source code is available at https://github.com/sebspel/DELight_reconstruction.

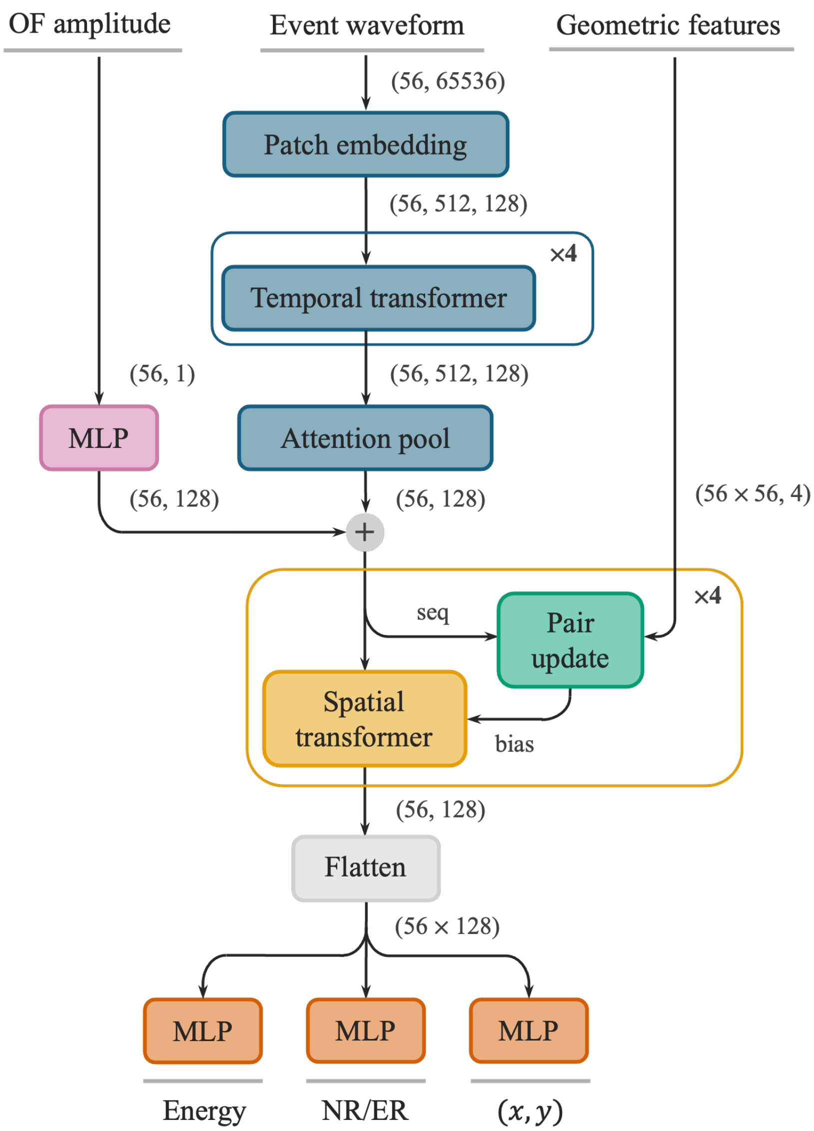

Figure 3.1: End-to-end DELiT pipeline. Each of the 56 sensor waveforms are segmented into non-overlapping patches and encoded by four weight-shared temporal transformer blocks with rotary position embeddings; attention pooling collapses the temporal axis to one embedding per sensor. The per-channel optimum-filter amplitude (§2.3) is projected by an MLP and added residually. Four spatial blocks then apply inter-channel self-attention biased by a learned pair representation, initialised from four geometric features per sensor pair and iteratively refined by a pair update. The 56 channel embeddings are concatenated and passed to three MLP heads for energy, NR/ER classification, and (x, y) position.

3.1 Data Preprocessing

The simulated dataset, comprising 300,000 events as defined in §2.2, is partitioned into training (70%), validation (10%), and test (20%) sets. Stratification is performed within each (energy, recoil type) pair. Events are assigned to partitions using a fixed shuffled order seeded with the global training seed (42). This ensures proportional representation and bitwise reproducibility. The same partitioning is used for both DELiT and the baseline (§3.2).

Input waveforms are not normalised. Per-event normalisation would rescale the absolute signal amplitude that encodes recoil energy, and per-channel normalisation would distort the cross-channel amplitude ratios that encode position. Reconstruction targets are also left in physical units (eV and mm), so losses and reconstruction errors remain directly interpretable. The differences in scale between tasks are addressed by the weighted loss described in §3.6. The z-coordinate is not reconstructed because the phase-I geometry does not have side-wall instrumentation and therefore does not offer vertical sensitivity.

To improve robustness to partial readouts, channel masking is applied during training. For each event, either the 37 above-liquid or the 19 submerged channels are zeroed as a coherent group with 10% probability each, mutually exclusively; the remaining 80% of events are unmasked. This is referred to as channel masking throughout. Mask draws are seeded with the global training seed (42); the masking pattern is deterministic, although downstream training is reproducible only within seed-level variance (§3.7). The mask is applied multiplicatively to both the waveform and any injected OF amplitudes. During inference, performance is evaluated under three configurations: all channels active, above-liquid channels only, and submerged channels only. This approach assesses the model's ability to sustain performance when an entire sensor layer is unavailable.

3.2 Optimum Filter Baseline

To quantify the benefit of joint temporal–spatial processing, a conventional baseline is established using gradient-boosted decision trees (BDTs), implemented with XGBoost [26]. The models are trained on the per-channel OF outputs described in §2.3. Gradient boosting constructs an ensemble of shallow decision trees in sequence, with each subsequent tree trained to correct the errors of the preceding model [26]. This approach is widely adopted in high-energy physics for classification and regression tasks involving fixed-length feature vectors [27].

For each of the 56 sensor channels, the OF yields an amplitude $\hat{a}$ and a goodness-of-fit χ^2. Concatenating these quantities across all channels results in a 112-dimensional feature vector for each event. Since BDTs do not inherently support multi-task learning, four separate XGBoost models are trained: a regressor for energy prediction, a binary classifier for NR/ER discrimination, and independent regressors for the x and y interaction coordinates.

The energy regressor is trained on log-transformed targets, ensuring that the squared-error loss penalises relative rather than absolute deviations. This approach emphasises low-energy events, where the fractional resolution is most critical. All four models share the same random seed (42) and hyperparameters: a maximum tree depth of 6, a learning rate of 0.03, and up to 2000 estimators, with early stopping implemented after 50 rounds without improvement in the validation loss. All remaining XGBoost hyperparameters are left at their library defaults. These settings are selected based on the dataset scale and are validated by observing stable convergence in both training and validation loss curves. An exhaustive hyperparameter search is not conducted due to limited access to batch-queued compute resources. All models use the data partitions defined in §3.1 and are evaluated using the metrics described in §3.8.

3.3 Temporal Processing

Temporal processing is applied independently to each channel. The same set of parameters is used for every channel, so the temporal stack learns pulse-physics features without entangling them with the channel's position in the detector. Inter-channel correlations are introduced separately in the spatial stage (§3.5).

Each sensor channel records a waveform of 65,536 samples, which is partitioned into 512 non-overlapping patches of 128 samples. Each patch is linearly projected to a $d_{\text{model}} = 128$ embedding dimension by a single bias-included linear layer (Figure 3.1). Each 33 μs patch is shorter than the dominant signal components, the quantum evaporation signal $\mathcal{O}(100\,\mu\text{s})$–$\mathcal{O}(1\,\text{ms})$ and the triplet excimer $\mathcal{O}(1\,\text{ms})$ [12], so substructure within each component is resolved across multiple patches and sequence length is reduced by a factor of 128.

Temporal information is encoded using Rotary Position Embeddings (RoPE), which apply position-dependent rotations to query and key representations [28]. This method encodes relative temporal offsets, enabling the model to capture relationships such as delays between scintillation and evaporation signals without relying on absolute indexing.

Embedded patches are processed by four pre-normalised transformer blocks with layer normalisation applied before each sub-layer and residual connections around them. Each block comprises a *gated multi-head self-attention* module followed by a two-layer MLP. The attention module uses four heads of per-head dimension 32, and its output is modulated element-wise by a learned sigmoid gate of the block input before the output projection [29]. The MLP employs SwiGLU activation [30], implemented as two parallel linear projections from d_{model} to an intermediate dimension of 512 whose outputs are combined by element-wise multiplication after a SiLU nonlinearity, followed by a projection back to d_{model}. Dropout of 0.15 is applied after attention and within the MLP. Linear projections in all attention modules use Xavier-uniform initialisation [31]; the remaining weights use PyTorch defaults.

Following temporal encoding, a learned attention pooling operation reduces the 512-position sequence to a single embedding per channel. The pooling uses the same gated multi-head attention module as the transformer blocks, but as cross-attention. A single learned query, shared across channels, attends to all 512 temporal positions, with RoPE and dropout disabled. Unlike fixed pooling schemes, this mechanism assigns adaptive weights across the sequence, emphasising informative regions such as signal onset or transitions between components while suppressing baseline noise.

The temporal stage produces an output tensor of shape (56, 128) per batch, corresponding to one embedding per channel per event. The model uses $d_{\text{model}} = 128$, four transformer blocks, and four attention heads—the largest configuration that fits within per-GPU memory at the device batch size specified in §3.7.

3.4 Optimum Filter Feature Injection

The pooled per-channel embeddings contain only information recoverable from the raw waveform by the temporal stack. In contrast, the optimum-filter amplitude $\hat{a}$ depends explicitly on the noise PSD $J(\nu)$ and the layer-specific template $A(\nu)$ (§2.3), neither of which is provided to the model. Since $\hat{a}$ is therefore not generally reconstructible from the waveform alone, it is injected as an auxiliary feature. By comparison, the goodness-of-fit χ^2 is a scalar summary of the trace–template residual, which the temporal transformer can infer directly from the waveform up to the unknown weighting by $J(\nu)$. Unlike $\hat{a}$, it introduces no fundamentally inaccessible information and is therefore not included.

Injection is applied after attention pooling rather than at the patch level. Patch-level concatenation would repeat the same per-channel scalar across all 512 temporal positions, adding no new information after the first attention layer while unnecessarily increasing the embedding dimension throughout the temporal stack. Post-pool injection instead adds the feature once per channel, exactly where the per-channel summary is formed.

The scalar $\hat{a}$ is projected to $d_{\text{model}} = 128$ by a two-layer MLP with GELU activation and added residually to the pooled embedding (Figure 3.1). The MLP maps the scalar into the same feature space as the temporal embedding and allows a non-linear dependence, rather than a fixed linear shift, which is useful where OF assumptions break down due to template mismatch (§2.3). Residual addition preserves the temporal embedding as the main signal pathway, while letting the network learn the relative weight given to $\hat{a}$. The OF amplitude is expected to help mainly with energy regression and less with NR/ER discrimination or position reconstruction, which rely more on inter-channel correlations than on a single per-channel scalar. This expectation is tested in the ablation of §4.1.

3.5 Spatial Processing

The spatial block operates on the set of 56 channel embeddings produced by §3.3 and applies inter-channel self-attention biased by an explicit pairwise representation of the sensor array. Its design is adapted from the recent AlphaGenome realisation of the AlphaFold2 architecture, which maintains both a sequence representation and a pair representation that iteratively update one another across blocks [32, 33]. The adaptation here replaces learned biological pair priors with fixed detector geometry and uses four blocks rather than the dozens used in protein and genomic models, reflecting the smaller token count ($N = 56$) and the absence of a coevolutionary signal.

Each sensor pair (i, j) is assigned four precomputed geometric features: the relative displacement components $(\Delta x_{ij}, \Delta y_{ij}, \Delta z_{ij})$ and a binary same-array indicator (above-liquid or submerged). The displacement is supplied as a vector rather than a scalar distance so that direction-dependent signal asymmetries remain accessible. Δz_{ij} is retained even though z is not a reconstruction target, because its sign distinguishes the two readout layers and enables the pair update to express layer-crossing relationships that the scalar same-array indicator cannot. These features are fixed by the detector layout and shared across all events.

The geometric features and channel embeddings are combined in the sequence-to-pair operation (Figure 3.2, left). Channel embeddings pass through a GELU nonlinearity and two distinct linear projections, producing a "left" representation for sensor i and a "right" representation for sensor j whose outer sum across all sensor pairs yields a $(56, 56, 128)$ tensor. The asymmetry of these projections preserves the directional information in $\Delta r_{ij} = -\Delta r_{ji}$, which a symmetric construction would erase. In parallel, single-head dot-product attention $q_i \cdot k_j/\sqrt{d_{\text{model}}}$ between channel embeddings is augmented by a scalar bias from the geometric features, computed by a two-layer MLP (hidden dimension 64). The resulting per-pair attention output is lifted to d_{model} by a learned linear projection and added to the outer-sum tensor. This construction makes the pair representation jointly dependent on the learned channel features and the fixed detector geometry, providing a base representation that subsequent pair updates can refine.

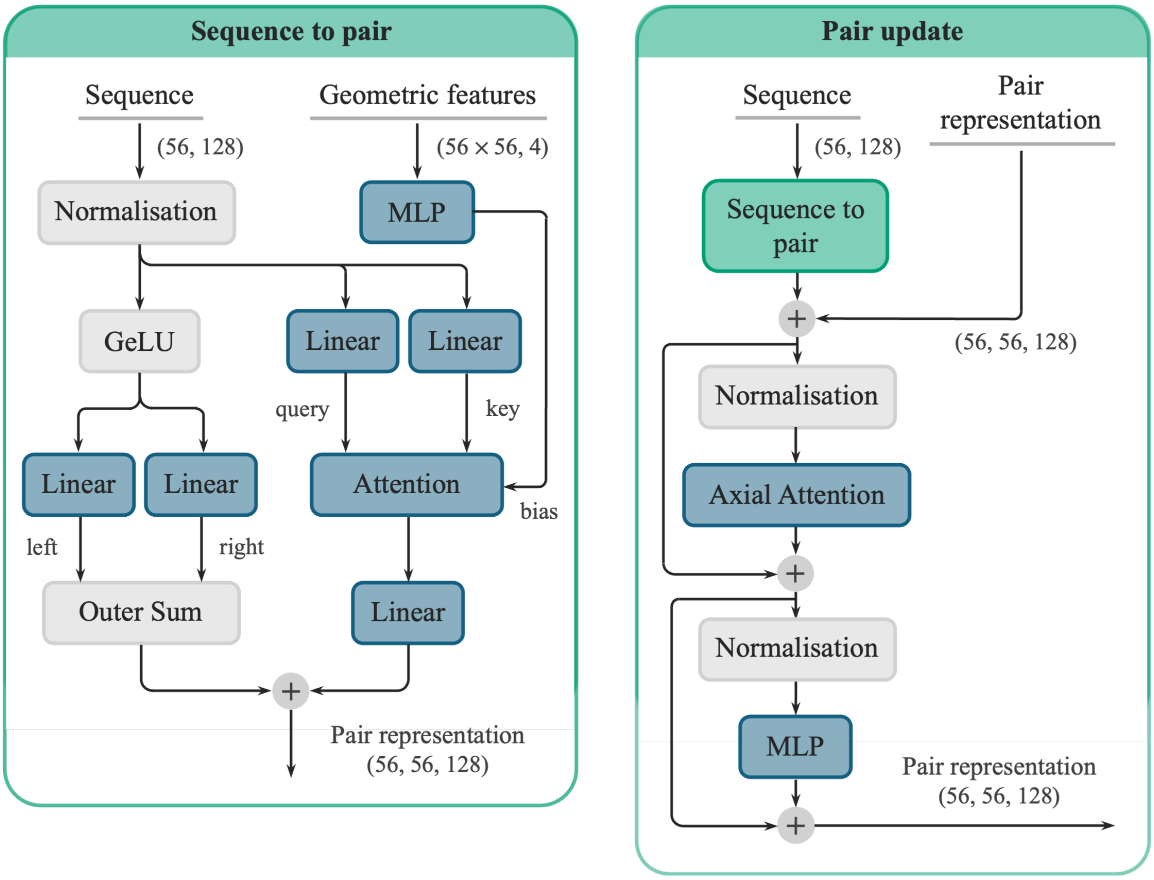

Figure 3.2: Detail of the spatial processing sub-components, adapted from the architecture of AlphaGenome [32]. The sequence-to-pair operation (left) constructs a pairwise representation from channel embeddings and precomputed geometric features. The pair update (right) refines the pair tensor through axial self-attention (row-wise followed by column-wise) and an MLP. Layer normalisation is applied before each operation. Residual connections are indicated by $\oplus$.

The pair tensor is iteratively refined by a pair update step (Figure 3.2, right). Axial self-attention is applied row-wise then column-wise on the $(56, 56, 128)$ tensor, followed by an MLP. The row-wise pass lets all pairs sharing a common source sensor i exchange information; that is, for each fixed i, the pair embeddings $(i, 1), (i, 2), \ldots, (i, 56)$ attend to one another. The column-wise pass performs the same operation on pairs sharing a target sensor j. Concretely, the row pass lets each sensor compare what it sees to all other sensors simultaneously, and the column pass then propagates those comparisons back so that every pair embedding (i, j) has access to the full set of i-relations and j-relations after two passes. This factorisation reduces the cost of full pair self-attention from $O(N^4 d)$

to $O(N^3 d)$ while still propagating information across the entire pair space in two passes. The refined pair tensor is projected to per-head scalars by a shared Linear($128 \rightarrow 4$) map, producing an attention bias of shape $(4, 56, 56)$ that is added to the dot-product logits of the spatial transformer layer before the softmax. Each spatial block consists of a pair update followed by a transformer layer of the same gated multi-head attention and SwiGLU configuration as the temporal blocks (§3.3). RoPE is disabled because the channel axis has no meaningful ordering, and inter-channel geometry is already provided by the pair-derived bias. Four such blocks are stacked, matching the depth of the temporal stack. The pair tensor is initialised to zero in the first block and updated residually, allowing the geometric prior to adapt progressively to each event's signal topology. The contributions of the pair update and the geometric features, both individually and jointly, are quantified in the ablation of §4.1.

3.6 Task Heads and Multitask Loss

The 56 updated channel embeddings are concatenated into a single 7,168-dimensional vector and passed to three independent two-layer MLP heads with hidden dimension 128 and GELU activation. The energy head outputs a scalar prediction $\hat{E}$ trained against the true recoil energy (eV) under mean squared error (MSE). The position head outputs two coordinates $\hat{\mathbf{r}} = (\hat{x}, \hat{y})$, trained against the true position (mm) under MSE. The classification head outputs a single logit, trained under binary cross-entropy with NR=1 and ER=0. The corresponding per-task losses are $\mathcal{L}_E$, $\mathcal{L}_{\mathbf{r}}$, and $\mathcal{L}_c$.

The targets are kept in their physical units (§3.1), so the three losses span a wide range at initialisation: $\mathcal{L}_E \approx 4.5 \times 10^4$ eV^2, $\mathcal{L}_{\mathbf{r}} \approx 5.8 \times 10^3$ mm^2, and $\mathcal{L}_c \approx 0.7$. An unweighted sum would therefore be dominated by $\mathcal{L}_E$, effectively starving the other heads of training signal. The losses are combined as a fixed weighted sum

$$\mathcal{L} = w_E \mathcal{L}_E + w_{\mathbf{r}} \mathcal{L}_{\mathbf{r}} + w_c \mathcal{L}_c, \qquad w_E = 10^{-4}, w_{\mathbf{r}} = 10^{-3}, w_c = 1, \tag{3.1}$$

with the weights chosen so that each term contributes comparably to the gradient at initialisation. The values were fixed by inspection of the raw loss magnitudes at step zero. No search was performed because (i) all reported ablations use an identical loss configuration, so any shift in absolute performance is common to every variant and the relative ranking is unbiased, and (ii) the available batch-queued compute did not permit a search. Adaptive weighting strategies such as uncertainty weighting [34] were not pursued; an evaluation of their effect on DELiT is left to future work.

3.7 Training Configuration

Models are trained with the AdamW optimiser [35]; the learning rate is linearly warmed up over 1,000 steps, then cosine-decayed to zero [36]. Each effective batch of 256 events is formed from 8-event microbatches with 32 gradient accumulation steps to fit single-GPU memory; one training step denotes one optimiser update. Training is step-bound at 10,000 steps (12.19 epochs of the 210,000-event training set) with no early stopping. Preliminary runs showed training loss converged and validation loss stable by this point, indicating no benefit from longer training. The validation set is used only to monitor overfitting and select model configurations for test-set evaluation; the test set is not used during development. Hyperparameters are listed in Table 3.1.

Table 3.1: Training hyperparameters. The same configuration is used for all DELiT variants in §4.1.

Hyperparameter	Value	Hyperparameter	Value
Learning rate	3×10^{-4}	Training steps	10,000
(β_1, β_2)	(0.9, 0.999)	Batch size	256 (8 micro $\times$ 32 accum.)
Weight decay	4×10^{-2}	Gradient clipping	Max norm 5.0
LR schedule	Cosine $\to$ 0	Dropout	0.15
Warmup steps	1,000	Channel mask probability	0.1
Random seed	42	Precision	bfloat16 mixed

The channel masking introduced in §3.1 is active only during training, applied multiplicatively to both input waveforms and, where present, injected OF amplitudes. Training is performed on a single NVIDIA A100-PCIE-40GB GPU in bfloat16 mixed precision with `torch.compile` enabled, and one full 10,000-step run takes approximately 36 hours.

The global seed (42) fixes data partitioning, channel-mask draws, parameter initialisation, and the dataloader's minibatch ordering. Bitwise reproducibility is not achievable because of non-determinism in CUDA kernels under mixed-precision compilation (PyTorch 2.5.1, CUDA 12.4), so reruns reproduce results within seed-level variance rather than bitwise.

3.8 Evaluation Metrics

Performance is evaluated on the full 60,000-event held-out test set under `torch.no_grad` in bfloat16 inference on the same A100 used for training, with dropout disabled. Each masking configuration from §3.1 is evaluated in 3,750 steps at batch size 16, taking approximately 12.7 minutes per configuration. Reconstruction quality is reported at the simulated energies defined in §2.2 using two complementary metrics per task; each value carries an uncertainty estimate matched to the statistical structure of the quantity.

Energy reconstruction is best characterised by its resolution and its bias

$$\frac{\sigma_E}{E} = \frac{\mathrm{std}(\hat{E})}{E}, \qquad \beta_E = \frac{\langle \hat{E} \rangle - E}{E}, \tag{3.2}$$

where both are evaluated within each energy bin, and $\mathrm{std}(\hat{E})$ is the sample standard deviation. The resolution is equivalent to the ΔE_{FWHM} reported in single-sensor MMC analyses [9], differing only by a factor of $2\sqrt{2 \ln 2}$. Reporting the fractional resolution rather than absolute σ_E matches the calorimetry convention of quoting an energy-resolving power E/σ_E [9] and allows the noise-limited and signal-limited regimes to be compared on a single axis. The pair $(\sigma_E/E, \beta_E)$ is reported because it cleanly separates stochastic resolution effects from systematic energy-scale bias; a model with strong resolution but a non-zero β_E would still distort the reconstructed recoil spectrum that downstream dark-matter analyses operate on.

Recoil classification is primarily characterised by the ER leakage fraction at 50% NR acceptance, the discrimination working point adopted by the DELight collaboration [12]. For each energy bin, the threshold p_{50} on the predicted NR probability is set so that exactly half of the true NR events are retained. The ER leakage fraction is then

$$f_{\text{leak}} = \frac{N_{\text{ER}}(p > p_{50})}{N_{\text{ER}}}, \tag{3.3}$$

where $N_{\text{ER}}(p > p_{50})$ is the number of true ERs whose predicted NR probability exceeds p_{50}. This directly maps onto background rejection at fixed signal efficiency. The area under the receiver operating characteristic curve (ROC-AUC) is reported as a secondary, threshold-independent measure of class separability, defined as the probability that a randomly selected NR receives a higher predicted NR probability than a randomly selected ER [37]. Both classification metrics depend only on the rank ordering of predicted scores within each class and are therefore invariant to the NR/ER ratio. As a result, evaluation on the balanced test set provides estimates that generalise directly to the strongly imbalanced ER-dominated regime in dark matter searches.

Position reconstruction is characterised by a scalar RMSE per bin and a radial bias profile

$$\text{RMSE} = \sqrt{\frac{1}{2N}\sum_{n=1}^{N}\left[(\hat{x}_n - x_n)^2 + (\hat{y}_n - y_n)^2\right]}, \qquad \beta_r = \langle r_{\text{pred}} - r_{\text{true}}\rangle, \tag{3.4}$$

where β_r is computed in bins of the true radial position $r_{\text{true}} = \sqrt{x^2 + y^2}$. RMSE exploits the cylindrical symmetry of the planar sensor array (Figure 1.1), under which the in-plane axes are statistically equivalent. It is, however, blind to coherent radial structure: a regressor trained by MSE on a bounded support tends toward the conditional mean of plausible reconstructions, displacing predictions inward near the boundary; β_r is reported to expose this effect.

Uncertainties on σ_E/E, RMSE, and ROC-AUC are estimated by non-parametric bootstrap resampling [38]. Test events within each energy bin are resampled with replacement 1,000 times, the statistic is recomputed on each resample, and the reported error bar is the standard deviation of the resulting distribution. Bootstrap resampling is used because it requires no assumption about the shape of the prediction-residual distribution and handles derived statistics (ratios, square roots, areas under curves) without analytic error propagation. The energy bias and radial position bias carry the standard error of the mean, $\sigma/\sqrt{N}$, as the appropriate uncertainty for a sample-mean estimator. The ER leakage fraction carries the binomial standard error $\sqrt{p(1-p)/N}$, the Wald normal approximation [39] to the Bernoulli variance, where p is the measured leakage and N the number of true ERs in the bin. At bins where no ERs pass the threshold, the Wald estimator collapses to zero and a one-sided 68% Clopper-Pearson upper limit [40] is reported instead. All bootstrap resamples use a fixed random seed equal to the training seed (42) for reproducibility.

The analysis threshold is defined as the lowest energy bin at which the multi-channel reconstruction resolves a signal at five standard deviations, $E/\sigma_E(E) \geq 5$. Unlike the 20 eV trigger threshold projected for DELight, which is set by the energy-independent baseline noise of the sensor [12], the analysis threshold depends on the energy-varying reconstruction resolution $\sigma_E(E)$ and captures both propagated detector noise and learned-model error. The effective low-energy reach of the experiment is set by the largest of trigger and analysis thresholds.

4 Results and Discussion

4.1 Architecture Ablations

Three architectural components distinguish DELiT's spatial processing from a plain transformer operating on the same 56-channel waveforms: (i) injection of the optimum-filter amplitude (§3.4); (ii) the pair update mechanism (§3.5); and (iii) the use of pairwise geometric features as an additive attention bias (§3.5). Each encodes a specific physics-informed inductive bias. To isolate their individual contributions, six model variants are trained in which each component is independently disabled while the remaining architecture is held fixed. This evaluates whether incorporating physics-informed architectural priors improves reconstruction performance as discussed in §1.3.

The six variants are defined in Table 4.1 by the binary state of each component:

- **OF = 0:** The optimum-filter amplitude is not injected. The MLP branch shown in Figure 3.1 is removed, and the pooled per-channel embeddings pass directly into the spatial stack without the residual addition.
- **PU = 0:** No pair representation is constructed. The pair update block (detailed in Figure 3.2) is removed entirely, including the sequence-to-pair construction. If geometric features are used, they are applied directly to the spatial transformer as a static attention bias via a two layer MLP.
- **GF = 0:** No geometric features are passed to the model. The four pairwise features are replaced with a zero tensor wherever they would otherwise be consumed.

Table 4.1: DELiT ablation variants. Parameter counts are reported for reference.

Variant	OF	PU	GF	# Params	Description
V1	1	1	1	6 834 456	Full DELiT model
V2	1	1	0	6 834 456	Pair update without geometric priors
V3	1	0	1	5 119 240	Static geometric attention bias
V4	1	0	0	5 117 700	Plain spatial self-attention with OF injection
V5	0	1	1	6 817 688	Full geometric machinery without OF injection
V6	0	0	0	5 100 932	Plain spatial self-attention without OF injection

All variants are trained with the loss weights of §3.6 under the hyperparameters in Table 3.1, so observed differences reflect architectural capacity rather than differential convergence. Evaluation here uses the all-channels-active configuration; masked-inference results are deferred to §4.2.4.

Performance is reported at 50 eV and 500 eV. The 500 eV anchor lies in the high-energy regime of the training data, where signal amplitudes are large and architectural differences resolve cleanly. The 50 eV anchor provides a stable low-energy comparison at which all six variants separate; lower energies bear on the experimental threshold and are addressed in §4.2.5.

Three metrics are reported per variant: the energy resolution σ_E/E, the ER leakage fraction f_{leak} at 50% NR acceptance, and the position RMSE. The maximum observed f_{leak} across all variants and both energies is $0.120 \pm 0.049\%$, with the majority of entries consistent with zero leakage; classification is therefore not architecture-limited at these energies, and f_{leak} is omitted from Table 4.2.

Table 4.2: Architecture ablation results at energies of 50 eV and 500 eV. Uncertainties are bootstrap standard deviations over the test set. The best value in each column is shown in bold.

Variant	σ_E/E [%]		RMSE [mm]	
	50 eV	500 eV	50 eV	500 eV
V1	$\mathbf{20.9 \pm 0.3}$	$\mathbf{2.68 \pm 0.25}$	$\mathbf{4.57 \pm 0.04}$	$\mathbf{2.88 \pm 0.02}$
V2	23.7 ± 0.2	3.52 ± 0.22	5.26 ± 0.05	3.28 ± 0.02
V3	22.8 ± 0.3	3.30 ± 0.24	5.41 ± 0.05	3.82 ± 0.02
V4	26.1 ± 0.2	3.45 ± 0.22	5.46 ± 0.06	3.33 ± 0.02
V5	23.5 ± 0.3	2.82 ± 0.25	4.97 ± 0.04	3.34 ± 0.02
V6	27.4 ± 0.3	3.26 ± 0.23	5.67 ± 0.05	3.86 ± 0.03

Adding both the pair update and the geometric features (V4→V1) reduces σ_E/E by 20% at 50 eV and 22% at 500 eV, and RMSE by 16% at 50 eV and 14% at 500 eV. Without OF (V6→V5), the same comparison yields a 14% σ_E/E reduction at 50 eV (500 eV difference is insignificant within bootstrap uncertainty), and RMSE reductions of 12% and 13% respectively. The geometric machinery, therefore, contributes independently of OF.

V3 ingests the same pairwise features as V1 but applies them only as a static attention bias. Adding the iterative pair update (V3→V1) reduces σ_E/E by 8% at 50 eV (the 500 eV 19% reduction is at the edge of bootstrap significance), and RMSE by 16% at 50 eV and 25%. The position improvement at 500 eV is the largest single-component effect in the ablation, consistent with the pair representation constructing event-specific spatial structure. Adding the geometric features only as a static bias (V4→V3) yields a 13% σ_E/E reduction at 50 eV (500 eV insignificant); the effect on position is mixed, with RMSE unchanged at 50 eV and degraded by 15% at 500 eV. A geometric prior that the network cannot adapt to the event topology is therefore insufficient on its own and can act as a poorly matched constraint.

OF injection (V6→V4) reduces σ_E/E by 5% at 50 eV (500 eV not significant), and RMSE by 4% at 50 eV and 14% at 500 eV. With the full geometric machinery (V5→V1), σ_E/E is reduced by 11% at 50 eV (500 eV not significant), and RMSE by 8% at 50 eV and 14% at 500 eV. OF injection therefore contributes most clearly to energy resolution at low energy and to position reconstruction at high energy.

In summary, V1 attains the best σ_E/E and RMSE at both reference energies. The dominant contribution is the joint pair update and pairwise geometric features, which matter most at 50 eV. At 500 eV the temporal stack alone is sufficient, with several differences falling within bootstrap uncertainty, and static geometric features can degrade position. Inter-channel structure, therefore, carries the most information when the per-channel signal is sparse. The full DELiT model is selected for further evaluation.

4.2 Final Model Performance

DELiT's performance on energy regression, NR/ER discrimination, and (x, y) reconstruction is evaluated on the held-out test set against the OF-BDT baseline of §3.2 and across the masking configurations of §3.1. Error bars for all plots are defined in §3.8.

4.2.1 Energy Regression

Figure 4.1 (left) shows the energy resolution of the final DELiT configuration and the OF-BDT baseline across the six training energies. Above 50 eV DELiT improves on the BDT by a factor rising from 1.4 at 50 eV to 8 at 500 eV, with resolution falling from 20.9% to 2.7%. The BDT resolution stays near 30% between 50 and 200 eV and recovers to 21.5% at 500 eV.

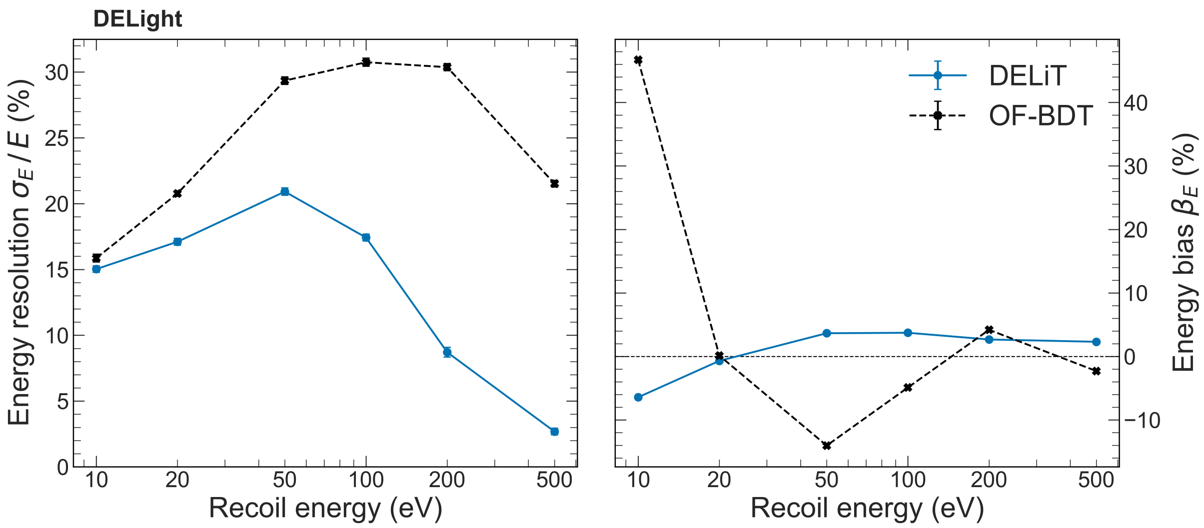

Figure 4.1: Energy resolution (left) and energy bias (right) as a function of true recoil energy. DELiT (blue) is trained on the 56-channel waveform data; the OF-BDT is trained on the features defined in §3.2.

Below 50 eV, both models show a non-monotonic response, with the resolution at 10 and 20 eV appearing better than at 50 eV. This reflects discretisation from the training energies: the model behaves more like a bin classifier than a continuous regressor, and the 10 and 20 eV results are not reliable indicators of continuous-energy performance.

The energy bias (Figure 4.1, right) tells a complementary story. DELiT remains within ±4% across the reliable range, while the BDT shows a +47% bias at 10 eV and oscillates between −14% and +4% above 50 eV. The 10 eV value, while not a reliable resolution point for either model, shows a distinct failure mode of the BDT. Its bias arises because low-energy events are harder to distinguish from noise and nearby higher-energy events, so predictions at the boundary tend to shift upward. A +47% bias near threshold would migrate events from below the analysis threshold to above it, producing a fake near-threshold excess.

The energy reconstruction advantage of DELiT above 50 eV arises because the per-channel OF summary used by the BDT (§3.2) discards fine-grained waveform structure. Pulse-shape deviations not captured by χ^2 and inter-channel amplitude correlations both carry additional energy information. By operating directly on the full 56-channel waveforms, the transformer blocks recover these features and convert them into improved resolution.

4.2.2 Recoil Discrimination

Figure 4.2 shows the ER leakage fraction at 50% NR acceptance (left) and the ROC AUC (right). Above 50 eV DELiT reaches sub-percent leakage and AUC > 0.995; the BDT shows $\sim$5% leakage and AUC 0.92 at 50 eV before converging toward unity at higher energies.

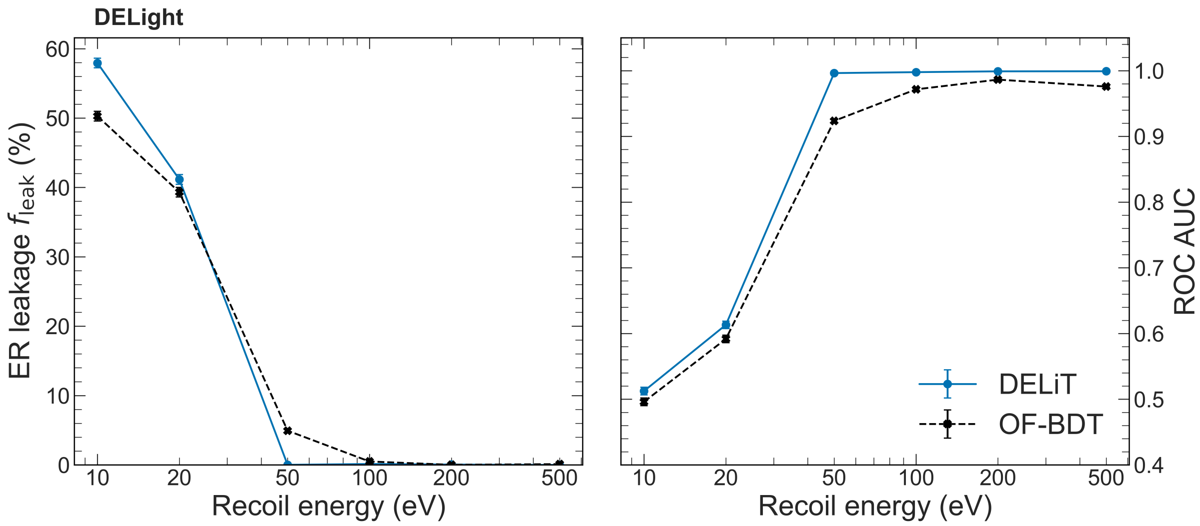

Figure 4.2: ER leakage at 50% NR acceptance (left) and ROC AUC (right) as a function of true recoil energy. DELiT (blue) is trained on the 56-channel waveform data; the OF-BDT is trained on the features defined in §3.2.

Both AUCs at 10 eV are $\sim$0.5, meaning that each model behaves as a random classifier. This is physically expected. The primary NR/ER discriminant is the quasiparticle-to-scintillation ratio, but below the 19.82 eV excitation threshold only quasiparticles are produced, so the discrimination power collapses. The apparent inversion in leakage (58% versus 50%) does not contradict this. Leakage at 50% NR acceptance is a single operating point on the ROC curve, whereas the AUC reflects performance across all thresholds. Although the AUC remains consistent with chance, small noise-driven deviations in the classifier output can shift slightly more ER events above a given threshold. These deviations are likely due to DELiT's higher parameter count fitting sample-specific noise structure that does not generalise, as expected when the class-conditional distributions are physically identical. At 20 eV the small differences (41% versus 39%, AUC 0.61 versus 0.59) reflect the same mechanism operating just above threshold: excimer production cross-sections remain small and physical discrimination information is sparse.

The advantage of DELiT at 50 eV and above has a clear physical origin. In the DELight geometry, scintillation produces signals in both the above-liquid and submerged layers, whereas quasiparticle-induced signals are confined to the above-liquid layer. The relevant discriminator is therefore an inter-channel amplitude asymmetry. Such cross-channel correlations are not efficiently captured by the BDT, whose per-channel feature construction does not naturally encode relationships between channels. In contrast, DELiT's spatial processing exploits the detector geometry by encoding pairwise features, while its temporal blocks resolve the separation between prompt scintillation photons and delayed quasiparticle evaporation signals.

4.2.3 Position Reconstruction

Figure 4.3 (left) shows the position RMSE of DELiT and the OF-BDT as a function of true recoil energy. Above 50 eV DELiT reaches 3–5 mm compared to 8–15 mm for the BDT, a factor-of-three improvement. Both models resolve position well below the 43.87 mm nearest-neighbour spacing of the above-liquid sensor array.

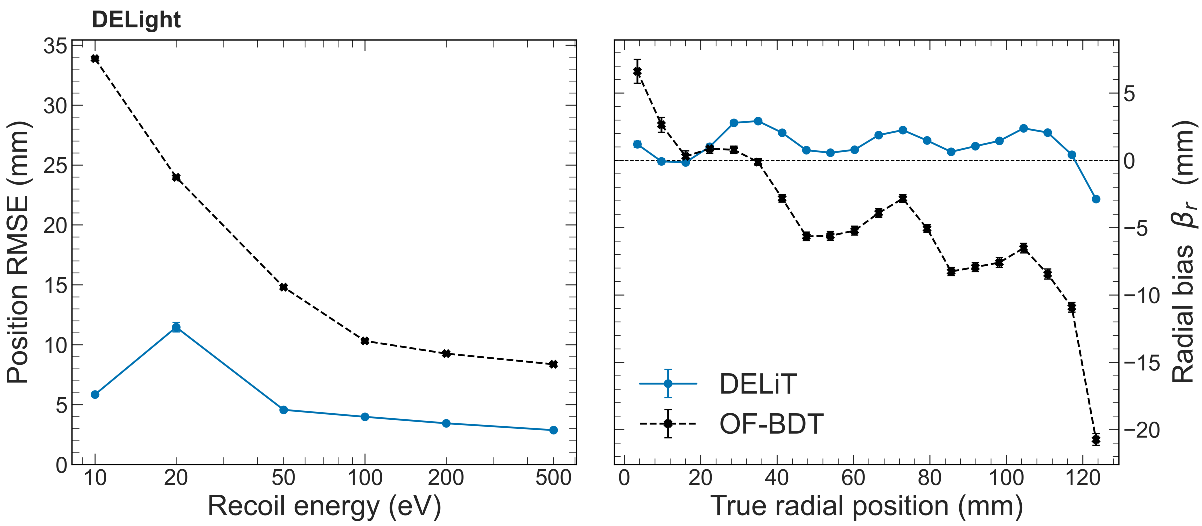

Figure 4.3: Position RMSE as a function of true recoil energy (left) and radial bias as a function of true radial position (right). DELiT (blue) is trained on the 56-channel waveform data; the OF-BDT is trained on the features defined in §3.2.

Both models exhibit a monotonic improvement in RMSE from 20 eV upwards, with the BDT maintaining this trend down to 10 eV at an RMSE of 33.9 mm. DELiT departs from this behaviour, attaining a substantially lower RMSE of 5.9 mm at 10 eV. The origin of this inversion lies in how each model uses spatial information in the quasiparticle-only regime below the 19.82 eV excitation threshold. In this regime, signal is confined exclusively to the above-liquid sensor layer, and position reconstruction must therefore rely on relative amplitude differences across the 37 channels. DELiT explicitly encodes pairwise geometric relationships between channels, enabling it to represent inter-channel amplitude comparisons as fundamental operations. This allows the model to specialise to this isolated regime. The BDT, by contrast, operates on scalar per-channel features with spatial identity implicit in feature ordering. This limits its ability to capture deeper relationships between channels.

DELiT's strong performance at 10 eV is therefore best interpreted as a dataset artefact: this energy point corresponds to a uniquely single-layer regime in the training data, and behaviour is not expected to generalise to continuous low-energy data.

The radial bias (Figure 4.3, right) shows DELiT is within ±3 mm across the fiducial range, while the BDT shows a pronounced inward bias, reaching −18 mm at the cell edge. This arises from the bounded training distribution: near the boundary, BDT predictions are effectively averaged over interior-supported samples, pulling estimates inward. DELiT does not exhibit this effect, as its geometric encoding allows near-wall signal patterns to map directly to near-wall positions.

4.2.4 Channel Masking

Figure 4.4 shows the energy resolution, ER leakage, and position RMSE under three inference configurations: all 56 channels active, 37 above-liquid channels only, and 19 submerged channels only. The masking protocol applied during training (§3.1) ensures that all three configurations are in-distribution for the trained model.

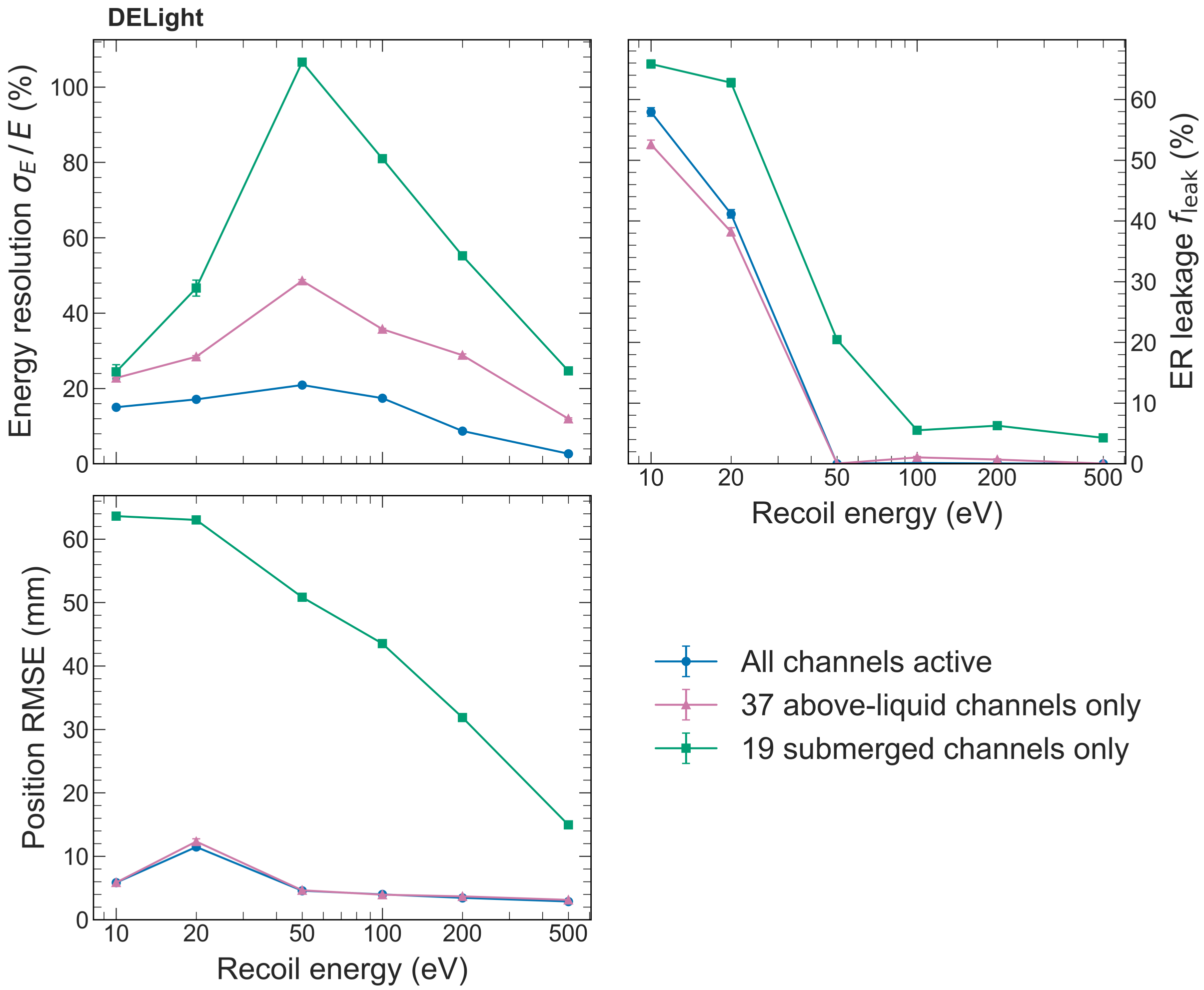

Figure 4.4: Energy resolution σ_E/E (top left), ER leakage at 50% NR acceptance (top right), and position RMSE (bottom left) as a function of true recoil energy for the three inference configurations described in §3.1.

At 10 and 20 eV, the above-liquid-only performance on all tasks reflects the same near-threshold effects discussed above for the all-channels baseline. Above 50 eV, the above-liquid-only energy resolution ranges from ∼49% at 50 eV to ∼12% at 500 eV, corresponding to a degradation factor of 2.3 at 50 eV and 4.4 at 500 eV relative to the full detector. The resolution still improves monotonically with energy.

ER leakage matches the all-channels result above 50 eV, reaching $< 1\%$ by 100 eV, while position RMSE remains within error bars of the all-channels curve at all energies. This is physically consistent: the 37 above-liquid sensors receive all three signal channels, sufficient for all reconstruction tasks. The energy resolution penalty therefore arises from the reduced total channel count rather than the loss of a distinct signal channel.

The submerged-only configuration is qualitatively different. At 10 and 20 eV, the energy resolution follows the above-liquid-only configuration with inflated values; at 50 eV it exceeds 100% and remains above 25% even at 500 eV, where the all-channels model achieves 2.7%. ER leakage follows the same shape as the other configurations; however, it stays above 20% at 50 eV and only falls to $\sim$5% at 500 eV. At 10 and 20 eV, position RMSE exceeds the 60.76 mm nearest-neighbour spacing of the submerged sensor array, improving to $\sim$15 mm at 500 eV. These large degradations follow directly from the loss of quasiparticle sensitivity in the above-liquid layer: less total energy is available to the model, the quasiparticle-to-scintillation ratio that drives NR/ER discrimination is inaccessible, and the spatial coverage from 37 sensor positions is removed.

DELiT has therefore learned a hierarchy of signal importance consistent with the underlying detector physics. The quasiparticle channel, carried exclusively by the above-liquid layer, is the dominant information source for all three tasks, and its removal produces degradation patterns that map directly onto the known signal transport asymmetry of the DELight geometry. The robustness of the above-liquid-only configuration also has a practical implication. If the submerged sensor layer failed or was excluded from a future analysis, the reconstruction pipeline would remain functional with degraded but usable performance across all three tasks.

4.2.5 Threshold Analysis

The effective low-energy reach of DELight depends on two distinct thresholds, each from a different stage of the detection pipeline. The hardware trigger is a two-level system [12]. The level-1 (L1) trigger operates on individual LAMCAL waveforms in an FPGA, applying an OF-based algorithm to distinguish signal pulses from baseline noise. Each LAMCAL has a design baseline resolution of 1–2 eV FWHM, corresponding to an L1 trigger threshold of approximately 4 eV per sensor at 4σ above baseline [12]. The level-2 (L2) trigger operates in software, combining L1 primitives (amplitude, timestamp, and channel identity) across the full 56-channel array to identify event-like timing patterns consistent with a particle interaction. Its design goal is an event-level trigger threshold of 20 eV [12]. This is a hardware quantity determined by MMC baseline noise and the trigger algorithm, independent of downstream reconstruction.

The analysis threshold is a reconstruction-level quantity, defined as the lowest energy at which the multi-channel reconstruction resolves a signal at $E/\sigma_E(E) \geq 5$. In this analysis, the 10 and 20 eV resolutions are not reliable indicators of continuous-energy performance (§4.2.1). The lowest reliable evaluation point is 50 eV, where DELiT achieves $\sigma_E/E = 20.9 \pm 0.3\%$, corresponding to $E/\sigma_E = 4.78 \pm 0.07$—narrowly below the 5σ criterion. At 100 eV, $\sigma_E/E \approx 9\%$ gives $E/\sigma_E \approx 11$, comfortably satisfying it. The analysis threshold therefore lies between 50 and 100 eV; the discrete training energies prevent interpolation between points. The 50 eV endpoint is adopted as the operating value: it is the lowest evaluation point consistent with the criterion at the level of bootstrap uncertainty, and it yields the strongest mass-reach claim available from the data. The effective low-energy reach is set by whichever threshold is higher: the analysis threshold of 50 eV exceeds the 20 eV trigger threshold, and the reconstruction is the binding constraint.

The analysis threshold can be mapped to a minimum detectable dark matter mass via the kinematics of elastic scattering [41, 42]. The maximum nuclear recoil energy for a dark matter particle of mass m_χ scattering off a ^{4}He nucleus of mass $m_{\mathrm{He}} = 3.727$ GeV is

$$E_R^{\max} = \frac{2\mu_{\chi N}^2\, v^2}{m_{\mathrm{He}}}, \qquad \mu_{\chi N} = \frac{m_\chi\, m_{\mathrm{He}}}{m_\chi + m_{\mathrm{He}}}, \tag{4.1}$$

where v is the dark matter speed in the laboratory frame. With the maximum halo speed $v_{\max} \approx 782$ km/s [43], the requirement $E_R^{\max} \geq E_{\mathrm{thr}} = 50$ eV yields $m_\chi \geq 121$ MeV. This situates DELiT's reach on the proposal's 50 eV sensitivity projection in Figure 1.2, showing that multi-channel reconstruction does not limit DELight's sensitivity to sub-GeV dark matter but imposes a stricter bound than the targeted 20 eV trigger threshold.

4.3 Limitations

The most consequential limitation of the threshold analysis is the use of six discrete training energies. DELiT is never exposed to intermediate energies between 10 and 50 eV, leading to the non-monotonic energy resolution and the anomalously low position RMSE in this regime (§4.2.1, §4.2.3). The network exploits the discrete label structure rather than learning a continuous energy–response mapping, producing optimistic estimates of σ_E/E and β_E below 50 eV and bounding the analysis threshold to the 50–100 eV interval rather than fixing a precise crossing point. The $\pm 4\%$ bias observed across the reliable range cannot be assumed to hold at untrained energies where the model must extrapolate.

All DELiT variants are trained from a single random seed. Stochastic variation in parameter initialisation and minibatch ordering can shift reconstruction metrics between otherwise identical runs [44]. In the ablation study (Table 4.1), several variants differ by 1–3% absolute in σ_E/E, which lies within plausible seed-to-seed variance for models of this scale. The component ranking is therefore indicative rather than statistically established, and multi-seed repetition would be required to assign confidence intervals to the ablation conclusions. Limited access to batch-queued compute prevented both multi-seed runs and systematic hyperparameter optimisation for DELiT and the OF-BDT, so the reported performance levels should be interpreted accordingly.

The dataset is entirely simulated. The simulation uses experimentally measured noise power spectral densities and physically motivated scattering cross-sections (§2.2), but assumes stationary Gaussian noise and perfect template fidelity for the signal waveforms. Real detector noise may exhibit non-stationarities, and pulse-shape deviations from the simulation template would propagate directly into the OF baseline and into DELiT through systematic mismatch between the simulated and real pulse-shape distributions. Validation against calibration data will be necessary once DELight is operational.

The training set uses a balanced 50:50 NR/ER ratio at each energy, whereas ERs dominate by orders of magnitude in a dark matter search. Both evaluation metrics—ER leakage at 50% NR acceptance and ROC AUC—are class-ratio invariant (§3.8) and quantify classifier separability, but do not fully capture performance in the low-ER-leakage regime. Since the model is optimised on balanced data, the absolute number of misclassified ERs passing the NR selection will scale with the true ER population, potentially requiring recalibration of the classifier operating point before use in an analysis pipeline.

5 Conclusion

This work developed and evaluated DELiT, a transformer-based architecture for joint energy regression, NR/ER classification, and (x, y) position reconstruction from DELight's 56-channel simulated waveform data, against the three aims of §1.3.

The first aim, joint multi-task reconstruction operating directly on the raw 65,536-sample waveforms, is met. DELiT delivers σ_E/E falling from 20.9% at 50 eV to 2.7% at 500 eV, sub-percent ER leakage above 50 eV, and position RMSE of 3–5 mm, improving on the OF-BDT baseline by factors of 1.4–8 in energy resolution and a factor of three in position RMSE while removing its inward radial bias near the detector wall.

The second aim, to test whether physics-informed architectural priors improve reconstruction, is answered affirmatively. The ablation of §4.1 isolates the pairwise geometric features and the iterative pair update as the operative mechanism, jointly reducing σ_E/E by 20% and position RMSE by 16% at 50 eV; the static geometric bias alone yields no consistent gain. OF amplitude injection contributes a smaller but consistent improvement. Channel masking corroborates these findings physically: the model assigns dominant importance to the above-liquid layer, the only one sensitive to quasiparticle evaporation, recovering the signal hierarchy imposed by the DELight geometry.

The third aim places the analysis threshold in the 50–100 eV interval under the $E/\sigma_E \geq 5$ criterion. Although narrowly missing the criterion at 50 eV ($E/\sigma_E = 4.78 \pm 0.07$), this energy is adopted as the operating threshold, giving the lowest mass-reach claim consistent with the bracket. Via the elastic-scattering kinematics of §4.2.5 this corresponds to $m_\chi \geq$ 121 MeV, a factor-of-two improvement on the 220 MeV reach set by HeRALD's 145 eV demonstrated threshold [11] and the lowest projected mass reach for any superfluid-helium dark matter detector. The reconstruction therefore preserves DELight's sensitivity to sub-GeV dark matter, though it currently sets the binding low-energy constraint, more restrictive than the projected 20 eV L2 trigger threshold [12].

Two directions follow from the limitations of §4.3 as natural extensions of this work. The first is to generate a denser low-energy dataset from the existing simulation framework, with multi-seed repetition to provide statistical confidence in the ablation ranking. This requires no architectural changes and is expected to better localise the threshold crossing point, potentially extending it below 50 eV. Secondly, the geometric encoding that produced the largest single ablation gain extends to the side-wall instrumentation planned for DELight phase-II, where three-dimensional displacements would provide a richer pair representation. It would also enable reconstruction of the z-coordinate, which is not possible in the phase-I dual-layer geometry used here. Adding z to the reconstructed (x, y) position fixes the full set of path lengths from the interaction site to every sensor, which determine the per-channel signal partitioning currently inferred by the model from amplitude patterns alone. Whether this translates into a lower σ_E/E at low energy, and therefore a lower analysis threshold, is an open question for phase-II data; the architecture is positioned to exploit such information if the dominant low-energy resolution limit is geometric rather than noise-driven.

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